DPS hardware for vertical drift

Existing BDE design, additions and procurement

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I&I workshop draft — 27 October 2026

DEEP UNDERGROUND NEUTRINO EXPERIMENT

Fermilab

Use a commercial PLC for VD detector protection

Proposal: use the bottom drift electronics (BDE) design and add the inputs and outputs needed by the other VD systems.

Put the central programmable logic controller (PLC) in the CUC, with distributed I/O near the equipment.

OPC UA carries DPS status and supervisory requests to and from slow controls.

CUC = Central Utility Cavern; central-PLC location proposed. HD can later use an extension or a separate PLC. This plan covers VD.

The VD rack plan already reserves space for DPS

Rack type	Count	DPS equipment / connections
BDE auxiliary	10	12U per rack; eight WIEC connections per rack
BDE low voltage	20	One PL506 per rack; four WIECs per supply
Bias high voltage	10	12U per rack for the DPS interface to ISEG supplies

The auxiliary racks contain the WIEC I/O; the bias racks contain the ISEG inhibit outputs.

May 2025 VD rack workbook; June 2026 BDE layout. U is rack height. These are design allocations; the bias-HV sheet is a first estimate.

One BDE I/O group serves eight WIECs

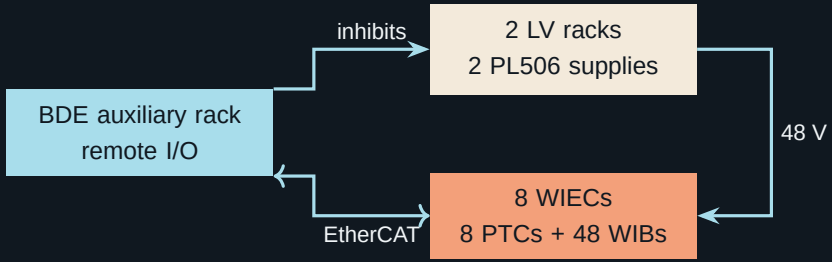


Diagram. Ten groups: 20 PL506 supplies, 80 WIECs and 480 WIBs.

The auxiliary rack also connects the fans, heaters and temperature sensors.

WIEC = warm interface electronics crate; PTC = power and timing card; WIB = warm interface board. June 2026 layout; fan/heater wiring omitted here.

Panel drawings and the Ignition front end are available

TPC protection panel designs

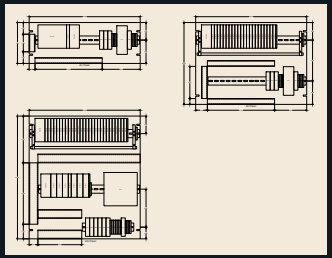
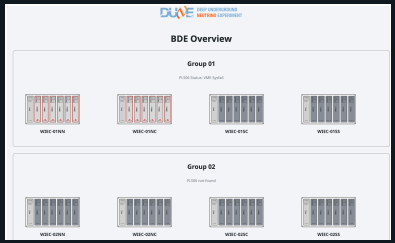


Diagram. Central PLC, bias I/O and LV/fan/heater panels.

BDE Ignition screens



Existing display of WIEC groups; June 2026 SC progress presentation.

Parts list: CX5140 PLC, EK1100 couplers, EK1521 fibre links and EL2124 inhibit outputs.

TPC reference: EDMS 2401090 v4, sheet 2; component names from the 2023 parts list.

The VD bias-panel draft specifies two DIN rails per rack

Component	Per rail	10 racks
EK1100 EtherCAT coupler	1	20
EL9410 supply terminal	1	20
EL2124: four 5 V outputs	18	360
EL9011 end cap	1	20
24 V power supply	1	20

Each rail occupies 337 mm. The draft provides 1,440 output channels; the working VD inventory lists 576 bias channels.

May 2025 workbook, Bias HV Rack!K17:K21. Quantities describe the first-pass panel layout.

VD power estimates for the BDE racks

Equipment	Per rack	VD basis
PL506 input	2.25 kW	20 racks: about 45.1 kW for the WIEC supplies
Auxiliary DPS input	About 1.11 kW	10 racks: about 11.1 kW; eight WIECs per rack
Bias DPS supply allowance	24 V, 40 A	960 W DC capacity per rack in the workbook

The auxiliary estimate scales the recorded 831 W from six to eight WIECs: $831 \times 8/6 = 1{,}108$ W.

Planning estimates from the May 2025 workbook. Linear scaling follows its VD note; it is not a measured load. Supply capacity is separate from AC demand.

The PX3 rack-protection setup already defines shutdown

June 2026 setup guide	Documented behaviour
Ignition connected	Ignition reads sensors and switches outlets over SNMP
Ignition unreachable	PX3 local rules handle smoke, water and temperature; the network-switch outlet stays on
After a fallback trip	Power is restored manually or through Ignition

This setup combines Ignition control with autonomous PDU fallback. The VD integration includes the PLC protection permits.

DUNE-doc-32655 v8, pp. 3–6 and 14–16. The PDU fallback is documented; the PLC permit integration is the proposed extension.

The VD inventory covers the other subsystem connections

Subsystem	Working inventory and interfaces
PDS readout / calibration	42 DAPHNE boards and 8 calibration boxes; 58 LV feeds from MPOD modules
Power over fibre (PoF)	80 laser boxes; 640 temperature and 640 voltage readings; compact PLC interface proposed
TDE	20 mezzanine racks; power supplies and readout-crate management
Calibration / cryogenic instrumentation	Level and temperature readout, purity monitors, ionization laser and neutron source

VD hardware lists and SC architecture. Device monitoring reaches Ignition through native drivers and OPC UA; protection uses the subsystem inhibit interfaces.

Protection timing includes detection and equipment shutdown

No critical protection function requiring a response below 1 ms has been identified in the current review.

Total response time includes the sensor, communication, PLC cycle, output and equipment shutdown.

The simulation exercises interface load and fault cases. Bench tests measure the complete hardware path.

The existing software benchmarks do not measure a PLC trip. Subsystem timing limits and the measured response form the acceptance criteria for each test.

DPS protection and SC monitoring have defined paths

Connection	Path in the design
PLC to detector equipment	EtherCAT remote I/O; PTC fibre links; wired supply inhibits
DPS to slow controls	OPC UA status, trip causes and supervisory requests; Ignition display and history
PDUs and power supplies	SNMP to the SC drivers; local device protection and DPS inhibit interfaces
Facility systems	Cryogenic controls via OPC UA; Metasys via BACnet in the SC drawing; limited SDSTA hardware exchanges proposed

BDE design, supplied SC architecture and the I&I correspondence. Proposed control rule: an SC operating request cannot override a protection trip.

The BMS drawing identifies 20 control-panel symbols

Drawing area	Interfaces on the inspection maps
CUC, level 4850	Eight panel references: air handling, chilled water, power and associated equipment
Level 4910	North and south sump interfaces, SP-1 and SP-2
4850-08-ES group	Spare panel for future equipment; mapped to the access-drift inspection area
Other 4850 areas	Chillers, pumps, ventilation and generator interfaces

Inspection planned with Rachel; Dale Curran is the SDSTA integration contact identified for this work.

U-FD-M-901 rev12, 14 February 2025; marked 4850/4910 maps and survey register. Symbols are drawing references; field observations are pending.

The VD bench tests use the BDE and G271 equipment

Test	Measure or demonstrate
PLC, remote I/O, PTC and supply inhibit	Complete trip time and equipment feedback
PDU sensors and shutdown logic	Response with SC connected, lost or stalled
SC commands, reset and re-enable	Protection takes priority; recovery is deliberate
Fieldbus, network and power failures	Output states, lost branches and restart behaviour

July G271 inventory: PDU, PL506, mini-MPOD and network switch installed; WIEC/PLC additions listed separately. The BDE vertical-slice work supplies the PTC/WIB context.

VD hardware work through the June 2027 review

Work	Deliverable
VD hardware adaptation	Panel and module allocation for the additional subsystem signals; CUC controller and fieldbus connections
Prototype tests	Trip-time measurements, fault response, SC/PDU integration and recovery results
June 2027 review	Drawings, parts and cable lists, power budget, costs, lead times and spares
Procurement / installation	Orders after readiness review; delivery and installation sequence for the site

Schedule basis: possible facility interfaces in summer 2027; full VD response-matrix review in early 2028. Existing documents support the hardware work now.

Hardware work and responsibilities

Proposed team	Work
Manuel + control engineer	VD hardware design, SURF interfaces and procurement package
Terri, Trevor + Manuel	BDE panels, PLC/I/O implementation and bench tests
Manuel + consortia	Equipment limits, subsystem signals and protective actions
Manuel + Geoff	SC/DPS command priority, PDU integration and recovery
Vivian + subsystem / DPS teams	ICD updates and interface records

Proposed division from the I&I discussion. Before DPS is available, Manuel proposes that each consortium remains responsible for protecting its equipment.

Reference: the May 2025 VD rack inventory

Rack type	Count	Rack type	Count
BDE low voltage	20	PDS low voltage	4
BDE auxiliary	10	Drift HV	2
TDE	20	Reserved drift HV	2
Bias HV	10	Spare / CalCI	8

76 numbered mezzanine positions: SEDM-01 to SEDM-76.

The worksheet also has a separate concept for 40 top-plate mini-racks.

May 2025 allocation: rack positions, distinct from PLC panels and remote-I/O stations. Includes the listed reserved and spare positions.